$I \leq 2$

$$\int \frac{1}{x(e^x+1)^\beta}\,dx \qquad x(e^x+1)^\beta \geq (2+x)^\beta$$

$$\geq x^{1+\beta}$$

$$\int \frac{1}{x^{1+\beta}} \quad (\forall \beta > 0)$$

$|a_n| \leq C$, $b_n \to 0$

$\forall \varepsilon > 0, \exists n_0 / \forall n \geq n_0, |a_n b_n - 0| \leq \varepsilon$. $f(b_n)$

$$|a_n b_n| = |a_n| * |b_n| \leq C|b_n|$$

$\varepsilon' = \varepsilon / C$. $\forall \varepsilon' > 0, \ldots |b_n| \leq \varepsilon' = C\varepsilon$

$\Leftrightarrow C|b_n| \leq \varepsilon$

$\Rightarrow |a_n b_n| \leq \varepsilon$ $\quad \forall \varepsilon$

$$\int_0^\infty \frac{1}{\mathrm{Log}(x)^2}\,dx \qquad \int_1^{1/2} \qquad \int_1^{1/2}$$

$$\lim \int_{-\alpha+\varepsilon}^{1} \frac{1}{x}\,dx + \int_{1+\varepsilon}^{2} \frac{1}{x}\,dx$$